

Abstract No.

Combined CDK9 and BET inhibition induces apoptosis in H3K27M pediatric diffuse midline gliomas through translational suppression and replication stress

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Diffuse midline glioma (DMG) harboring the H3K27M mutation is one of the most aggressive pediatric high-grade gliomas (pHGGs), with a dismal prognosis and no effective targeted therapies. The H3K27M mutation disrupts histone H3 methylation and acetylation, leading to widespread epigenetic dysregulation that alters gene expression and protein translation. Here, we sought to identify a targeted combination therapy exploiting these disease-specific vulnerabilities.

An epigenetic drug library screen, followed by combination and synergy screening, identified the combination of a CDK9 inhibitor (CDK9i) and a BET inhibitor (BETi) as the most potent therapeutic strategy for H3K27M DMG. High-content imaging of 3D tumoroids derived from H3K27M, H3 wild-type (WT), and H3G34V pHGG models, together with a non-malignant control, demonstrated significantly greater cytotoxicity of the CDK9i-BETi combination in H3K27M DMG than in the other models. Furthermore, the combination showed additive or synergistic effects with ionizing radiation, supporting its integration with current standard-of-care treatment.

Mechanistically, CDK9 inhibition suppressed translation and reduced MCL1 expression, while BET inhibition downregulated the expression of MYC, and E2F and MYC target genes. Combined treatment selectively induced replication stress in H3K27M cells, as evidenced by increased RPA32 phosphorylation, whereas this effect was markedly less pronounced in H3 WT models. These molecular changes culminated in robust apoptosis in H3K27M DMG cells in vitro. In vivo, the combination significantly reduced tumor burden in zebrafish early larvae xenografted with H3K27M DMG cells but showed limited efficacy against H3 WT and H3G34V tumors.

Together, these findings demonstrate that combined CDK9 and BET inhibition selectively targets H3K27M DMG by suppressing translation and inducing replication stress, ultimately triggering apoptosis. This therapeutic strategy warrants further investigation as a promising candidate for clinical translation in children with H3K27M DMG.



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